

03 AOU Test Plan

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AOU_CORE — Test Plan

1. Unit bench (aou/test/aou_core_tb.cpp)

Dual aou_core with connect_peer, FDI tied high, peer memory on B.

#	Case	Checks
1	CSR reset	ip_version, aou_con0, DISABLED in aou_init
2	APB decode errors	Bad length/alignment → TLM_GENERIC_ERROR_RESPONSE; offset ≥ kWindowSize → TLM_ADDRESS_ERROR_RESPONSE
3	Named CSR round trip	All-ones write/read-back on aou_interrupt_mask, lp_linkreset, prior_rp_axi, prior_timer matches each register's write mask
4	Per-RP CSR bank	Read/write at OFF_RP0_BASE (array-indexed decode path, outside every named offset)
5	AXI deny	Write while DISABLED → TLM_COMMAND_ERROR_RESPONSE
6	Activate	Local activate_start → both ENABLED; IRQ then W1C
7	Soft reset while enabled	aou_con0.aou_sw_reset → all CSRs + DISABLED on A only; B unaffected
8	AXI loopback	A axi_s[0] write/read appears in B memory
9	AXI bridge dest_rp out-of-range	dest_rp pointing past peer rp_count → TLM_ADDRESS_ERROR_RESPONSE
10	Deactivate	deactivate_start → DISABLED
11	aou_con0 WO paths	ip_version write is a no-op; deactivate_force (bit 0) forces ENABLED → DISABLED unconditionally

Pass: prints ALL TESTS PASSED, exit 0.

```
cd aou && ./run_tests.sh
./run_tests.sh --coverage # ≥ 95% on aou_core.cpp (currently 100%)
```

2. Platform smoke (sw/smc-vp-tests/smc-aou-test)

Hart-0 MMIO through smc-vp at SMC_AOU_BASE (0xC000_4000):

#	Case
1	Reset values (ip_version, aou_con0, dest_rp, DISABLED)
2	Activate → ENABLED + int_activate_start; W1C clear
3	Deactivate → DISABLED

```

cd sw/smc-vp-tests && ./run_smc_vp_tests.sh smc-aou-test

## 3. SMU platform (`sw/smu-vp-tests/smu-aou-ext-test`)

Both CPUs, after `aou_init.activate_start`:

| # | Check |
|---|-----|
| 1 | SMC write/readback of catch-all `0xA000_1000` through `xbar.ext_out` → AOU → peer |
| 2 | SEP reads AOU `ip_version` at `0x4000_4000` (dedicated SEP→SMC window) |
| 3 | SEP programs outbound filter 0, then write/readback `0xA000_1008` through
    `smn_outbound` → xbar → AOU |

```bash
cd sw/smu-vp-tests && ./run_smu_vp_tests.sh smu-aou-ext-test

```

...

### 3. Pass criteria

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- Unit: ALL TESTS PASSED
- Coverage: `./run_tests.sh --coverage` ≥ 95% line on touched sources
- ASan: `./run_tests.sh --asan` when `libasan` is available
- VP: firmware test prints ALL TESTS PASSED under `smc-vp`